BENTA Disease (B-cell Expansion with NF-κB and T-cell Anergy): A Comprehensive Disease Characterization Report

Target disease: BENTA Disease — *B-cell Expansion with NF-κB and T-cell Anergy* **Causal gene: CARD11** (CARMA1; HGNC:16393, NCBI Gene 84433, UniProt Q9BXL7, chromosome 7p22.2) **Identifiers:** OMIM 616452 · MONDO:0014645 **Inheritance:** Autosomal dominant (germline gain-of-function); one biallelic case reported **Category:** Mendelian inborn error of immunity

Summary

BENTA disease is a rare, autosomal-dominant inborn error of immunity caused by germline gain-of-function (GOF) missense (and rare in-frame deletion) variants in *CARD11* — the gene encoding the lymphocyte scaffold protein CARMA1. CARD11 normally transmits antigen-receptor signals from the B-cell and T-cell receptors to the transcription factor NF-κB via the CARD11–BCL10–MALT1 (CBM) signalosome. In BENTA, the mutant CARD11 protein self-aggregates and nucleates a constitutively active CBM signalosome, driving **antigen-independent NF-κB activation**. The clinical consequence is a congenital, polyclonal, naïve-predominant **B-cell lymphocytosis** accompanied by **splenomegaly, lymphadenopathy, and recurrent infections**, with paradoxical **T-cell anergy** (functional hyporesponsiveness) despite the activating lesion.

The natural history is strikingly variable. Many patients are asymptomatic or improve without targeted therapy, while a subset suffers early-childhood fatal complications — in a 2025 systematic review of 34 cases, all six deaths occurred before age three, attributable to refractory hemophagocytic lymphohistiocytosis (HLH) or opportunistic infection. Across the reported literature (fewer than ~40 patients worldwide), HLH emerged in roughly 19% of cases, and B-lymphocytosis — though characteristic — was not universal. BENTA patients carry an **elevated lifetime risk of lymphoid malignancy**, spanning both B-cell lymphoma (the same GOF lesions occur somatically in activated B-cell diffuse large B-cell lymphoma, ABC-DLBCL) and, unexpectedly, T-cell lymphoma (subcutaneous panniculitis-like T-cell lymphoma, SPTCL, in two unrelated patients).

There is **no established curative therapy**. Management is supportive and immunomodulatory — corticosteroids, the mTOR inhibitor sirolimus, and the calcineurin inhibitor tacrolimus have been used with clinical benefit; splenectomy paradoxically worsens lymphocytosis and is avoided. Mechanism-based agents targeting the downstream CBM–NF-κB axis (MALT1 paracaspase inhibitors, BTK inhibitors), developed for ABC-DLBCL, represent rational investigational options. This report synthesizes eight confirmed findings from 21 primary papers across all 15 requested disease-characterization domains, distinguishing human clinical, model-organism, and in-vitro evidence throughout.

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1. Disease Information

Overview. BENTA (*B-cell Expansion with NF-κB and T-cell Anergy*) is a rare primary immunodeficiency / lymphoproliferative disorder — more precisely an inborn error of immunity with immune dysregulation — caused by germline gain-of-function mutations in **CARD11**. The hallmark is a congenital or early-childhood polyclonal expansion of B lymphocytes driven by constitutive NF-κB signaling, coupled with functional T-cell anergy. It was first defined molecularly by Snow et al. in 2012 (PMID: 23129749).

Key identifiers.

Resource	Identifier
OMIM	616452
MONDO	MONDO:0014645
Gene (HGNC)	CARD11, HGNC:16393
NCBI Gene	84433
UniProt	Q9BXL7
Cytogenetic locus	7p22.2
MeSH	No dedicated BENTA term; indexed under Primary Immunodeficiency Diseases / Lymphocytosis
ICD-10/ICD-11	No specific code; grouped with primary immunodeficiencies
Orphanet	Rare immune dysregulation / lymphoproliferation (no large registry entry)

Synonyms / alternative names. B-cell Expansion with NF-kB and T-cell Anergy; CARD11 gain-of-function disorder; CARD11 GOF; congenital B-cell lymphocytosis due to CARD11.

Source of information. The information base is a mix of **individual patient case reports and small case series** (the large majority of the literature) aggregated into two systematic reviews (Nishinosono et al. 2025, PMID: 39998705; Baldini-type review 2025, PMID: 40157432). There are no EHR-scale or population-registry datasets given the disease's rarity.

2. Etiology

Primary cause — genetic. BENTA is a **monogenic** disease caused by germline heterozygous (rarely biallelic) **gain-of-function variants in CARD11**. Snow et al. identified a germline heterozygous missense mutation **E127G** in an affected family and an independent **G116S** in an unrelated, phenotypically similar patient; both map to the CARD11 coiled-coil domain and drive constitutive NF-κB activation (PMID: 23129749):

> *"We found a novel germline heterozygous missense mutation (E127G) in affected patients in the gene encoding CARD11... We subsequently identified a second germline mutation (G116S) in an unrelated, phenotypically similar patient, confirming mutations in CARD11 drive disease."*

Genetic risk factors. The causal variants *are* the risk factors — this is a highly penetrant Mendelian disorder, not a polygenic/susceptibility trait. No separate GWAS susceptibility loci apply.

Environmental risk / trigger factors. Environmental exposures do not *cause* BENTA, but they **modulate severity and precipitate complications**. The most important is **Epstein–Barr virus (EBV) infection**, which markedly exacerbates B lymphocytosis; **splenectomy** similarly worsens the lymphocyte count and is therefore contraindicated (PMID: 25352053). Infections (including opportunistic) are major triggers of fatal outcomes in infancy.

Protective factors. No genetic or environmental protective alleles are established. Milder phenotypes track with specific hypomorphic GOF alleles (e.g., C49Y, K215del) rather than with modifier genes.

Gene–environment interactions. The clearest interaction is **CARD11-GOF × EBV**: the expanded, NF-κB-primed B-cell compartment responds to EBV with dramatic further expansion and heightened HLH risk. This is a mechanistic inference supported by longitudinal observation rather than controlled study.

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3. Phenotypes

BENTA phenotypes span clinical signs, physical manifestations, laboratory abnormalities, and malignancy predisposition. Core features and suggested HPO terms:

Phenotype	Type	Characteristics	Frequency	HPO suggestion
Polyclonal B-cell lymphocytosis	Lab abnormality	Congenital/early onset; counts 20,000–90,000/mm³ in severe cases; persistent	Characteristic but **not universal**	HP:0012132 (B lymphocytosis) / HP:0001974 (Leukocytosis)
Splenomegaly	Clinical sign	Early childhood; variable	Common	HP:0001744
Lymphadenopathy	Clinical sign	Early childhood	Common	HP:0002716
Recurrent infections	Clinical	Esp. respiratory; opportunistic in severe cases	Common	HP:0002719
T-cell anergy / hyporesponsiveness	Lab/ functional	Reduced antigen-receptor responsiveness; near-normal counts & Vβ diversity	Characteristic	HP:0005435 (Impaired T-cell function)
Hemophagocytic lymphohistiocytosis (HLH)	Clinical	Fever, splenomegaly, hypercytokinemia; can be fatal	~18.8%	HP:0003261-adjacent; HP:0012047 (Histiocytosis)
B-cell / T-cell lymphoma	Malignancy	B-cell lymphoproliferation; SPTCL in 2 patients	Elevated lifetime risk	HP:0012191 (B-cell lymphoma)
Autoimmune-like manifestations	Clinical	Polyarthritis, encephalitis (one maternal case)	Rare	HP:0001369 (Arthritis) / HP:0002383 (Encephalitis)
Adenotonsillar/ lymphoid hyperplasia	Physical	Airway obstruction in one patient	Rare	HP:0040286 (Abnormal tonsil morphology)

The core clinical presentation is summarized by the 2025 review (PMID: 40157432):

> *"typically presenting with early-onset polyclonal B-cell lymphocytosis, splenomegaly, lymphadenopathy, and recurrent infections"*

Severity and progression. Highly **variable expressivity**: mild alleles (C49Y, K215del) retain near-normal counts and vaccine responses, while severe cases (e.g., G123D, M365K, biallelic R331P) show massive lymphocytosis and life-threatening complications. The disease is generally chronic/lifelong; lymphocytosis is persistent rather than remitting.

Quality of life. No formal EQ-5D/SF-36/PROMIS data exist. Impact ranges from negligible (asymptomatic carriers) to severe (infants with HLH, airway-obstructing lymphoid hyperplasia, malignancy). Chronic infection susceptibility and surveillance burden affect daily functioning in symptomatic patients.

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4. Genetic / Molecular Information

Causal gene. *CARD11* (Caspase Recruitment Domain family member 11), a multidomain cytoplasmic scaffold expressed predominantly in hematopoietic/lymphoid tissue. Disease OMIM 616452.

CARD11 allelic series — three distinct Mendelian diseases from one gene:

Mechanism	Disease	OMIM	Direction
Biallelic loss-of-function	CARD11-deficiency combined immunodeficiency (CID with atopy)	615206	LOF
Heterozygous dominant-negative LOF	CADINS / CARD11-associated atopy (severe atopic dermatitis, infections)	617638	DN-LOF
Heterozygous gain-of-function	**BENTA**	**616452**	**GOF**

CARD11 is required for normal activation of NF-κB, JNK, and mTOR during antigen-receptor signaling (PMID: 34341167).

Reported pathogenic BENTA variants (allelic series):

Variant	Zygosity	Domain	Notes	PMID
E127G	Het	Coiled-coil	Founding variant, familial	23129749
G116S	Het	Coiled-coil	Founding variant, unrelated	23129749
G123D	Het (de novo & inherited)	Coiled-coil	Familial (mother→son); maternal autoimmune features	25352053 / 36129242
G123S	Het	Coiled-coil	—	reviews
C49Y	Het	CARD	Mild phenotype, 3 patients	25930198
M365K	Het	—	Mouse knock-in model	36960072
D357E	Het	—	—	40157432
K215del (in-frame 3-bp del)	Het	Coiled-coil	Mild, subtle NF-κB increase	31897776
R331P	**Homozygous (biallelic)**	—	B-cell reprogramming	41456721

Variant classification. Pathogenic / likely pathogenic per ACMG/AMP, supported by functional NF-κB assays. All are **missense or in-frame deletion** (no truncating variants cause GOF).

Allele frequency. Germline GOF variants are **absent or vanishingly rare in gnomAD**, consistent with pathogenicity and strong purifying selection.

Somatic vs germline. BENTA is defined by **germline** variants. Notably, the **same lesion class occurs somatically** in ABC-DLBCL — a key link between the Mendelian disease and lymphomagenesis.

Functional consequence. Gain of function — constitutive, antigen-independent NF-κB activation. The variants promote self-aggregation of CARD11 and spontaneous CBM signalosome assembly.

Modifier genes / epigenetics / chromosomal abnormalities. No specific modifier genes, epigenetic signatures, or large-scale chromosomal abnormalities have been established for BENTA. B cells show an aberrant transcriptional program (see §6), but no disease-specific methylation studies exist.

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5. Environmental Information

- **Environmental / toxic factors:** None causative. No occupational, radiation, or pollution associations.
- **Lifestyle factors:** None established.
- **Infectious agents: EBV is a key disease modifier/trigger**, markedly exacerbating B lymphocytosis and predisposing to HLH (PMID: 25352053). Opportunistic infections are major causes of early-childhood mortality. Infections do not *cause* BENTA but act on the genetically primed immune compartment.

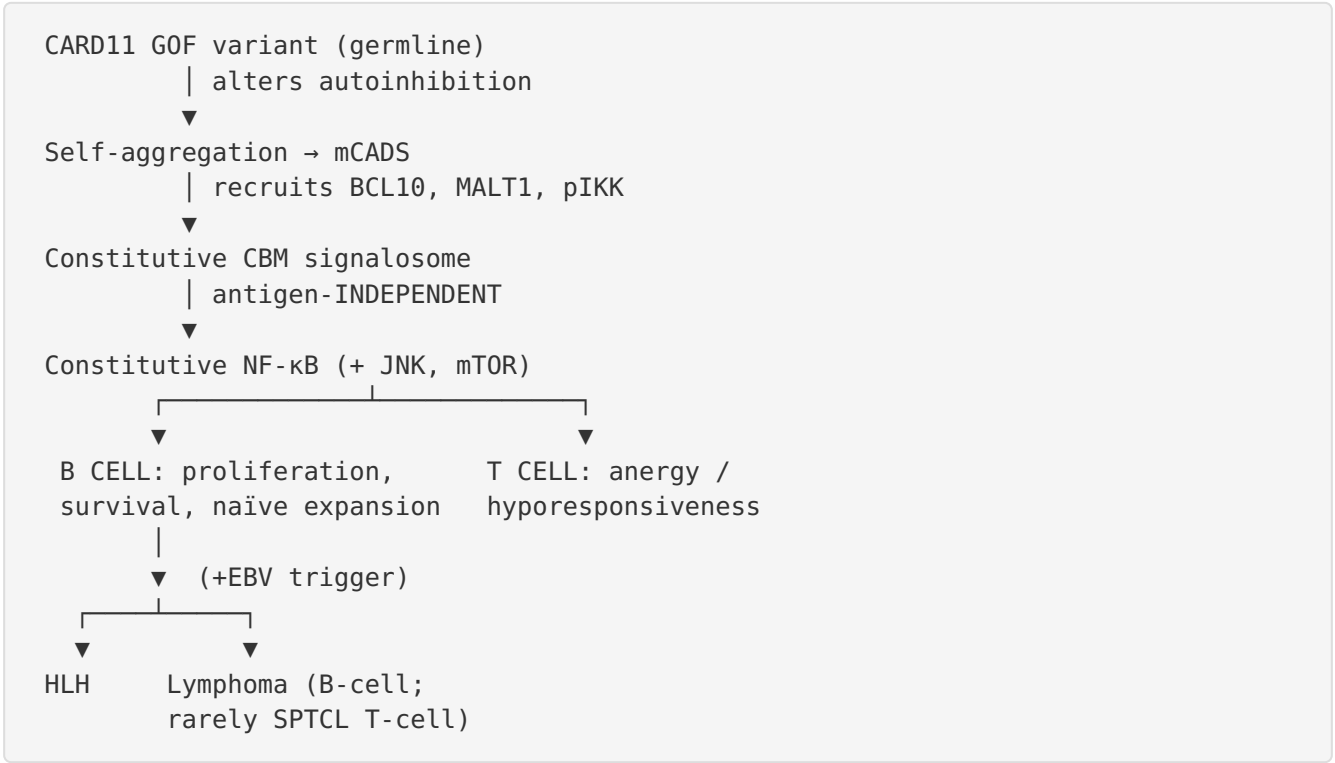
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6. Mechanism / Pathophysiology

Ordered causal chain

1. A **germline GOF variant in CARD11** (e.g., E127G, G116S, G123D, M365K, biallelic R331P), most often in the coiled-coil/CARD region, **alters the autoinhibited conformation** of the CARD11 scaffold.
2. This **leads to spontaneous self-aggregation** of mutant CARD11 into large cytosolic structures ("mCADS" — mutant CARD11-dependent shells) *[in-vitro demonstrated]*.
3. The aggregates **nucleate assembly of the CARD11–BCL10–MALT1 (CBM) signalosome** and **recruit phospho-IKK and MALT1**, i.e., an active signaling platform — **without antigen-receptor engagement**.
4. Constitutive CBM/IKK activity **results in antigen-independent, constitutive NF-κB activation** (with parallel JNK and mTOR signaling) *[demonstrated in reporter and primary-cell assays]*.

5. In **B cells**, constitutive NF-κB **drives upregulation of cell-cycle progression and survival genes**, **leading to enhanced proliferation and survival** and a **polyclonal, naïve-predominant B-cell expansion** (lymphocytosis, splenomegaly, lymphadenopathy).
6. In **T cells**, the same lesion **results paradoxically in functional anergy/ hyporesponsiveness** (reduced IFN-γ-producing $CD3^+$ cells), while counts and Vβ diversity remain near normal *[mechanism partly inferred — likely chronic-signaling-induced desensitization]*.
7. The expanded, NF-κB-primed compartment, further **triggered by EBV infection**, **branches toward HLH** (hypercytokinemia, impaired NK degranulation) and/or **toward lymphoid malignancy** (B-cell lymphoma; rarely SPTCL T-cell lymphoma), the leading causes of severe morbidity/mortality.



Mechanistic detail

Molecular pathways. Canonical **NF-κB** is the central effector (CBM → IKK → IκB degradation → NF-κB nuclear translocation). JNK/AP-1 and mTOR are co-activated. Suggested pathway references: KEGG NF-κB signaling; Reactome CBM-mediated NF-κB activation.

Protein dysfunction — gain of function via aggregation. Stinson et al. showed GOF CARD11 mutants form cytosolic aggregates that colocalize with the signaling machinery (PMID: 32473470):

> *"GOF CARD11 mutants can form large, peculiar cytosolic protein aggregates we term mCADS (mutant CARD11 dependent shells). MALT1 and phospho-IKK are reliably colocalized with mCADS, indicative of active signaling."*

and that these mutants *"drive constitutive NF-κB activity without AgR stimulation."* The CBM complex is the essential molecular hub (PMID: 25087226):

> *"The CBM complex forms an essential molecular link between the triggering of cell-surface antigen receptors and nuclear factor κB activation."*

Cellular processes. Dysregulated cell-cycle progression, enhanced survival/anti-apoptosis, and lymphoproliferation in B cells; T-cell anergy. Brohl et al. documented the B-cell program (PMID: 25352053):

> *"this patient's B cells exhibited higher expression of several cell cycle progression genes, as well as enhanced proliferation and improved survival following B cell receptor stimulation."*

Immune system involvement. A hybrid of **immune dysregulation (B-cell overgrowth), partial immunodeficiency (infection susceptibility, T-cell anergy), and hyperinflammation (HLH)**.

Transcriptomic / single-cell profiling. Bhattad et al. profiled a biallelic R331P patient (PMID: 41456721):

> *"massive expansion of B cells with a naive surface phenotype and aberrant transcriptional program."*

Suggested ontology terms. GO:0051092 (positive regulation of NF-κB transcription factor activity); GO:0038095 (BCR signaling pathway); GO:0002376 (immune system process); CL:0000236 (B cell); CL:0000788 (naive B cell); CL:0000084 (T cell); GO:0005737 (cytoplasm / aggregate localization).

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7. Anatomical Structures Affected

Organ level. Primary: **spleen (splenomegaly; UBERON:0002106), lymph nodes (lymphadenopathy; UBERON:0000029), peripheral blood (UBERON:0000178), bone marrow (UBERON:0002371)**. Secondary: **liver (hepatomegaly in HLH; UBERON:0002107)**, **tonsils/adenoids (lymphoid hyperplasia; UBERON:0002372)**, skin (SPTCL; UBERON:0002097). Body system: **hematopoietic / immune system (UBERON:0002405 immune system; UBERON:0002390 hematopoietic system)**.

Tissue / cell level. Lymphoid tissue; the expanded population is **naïve B cells (CL:0000788 naive B cell / CL:0000236 B cell)**; T cells (CL:0000084) are functionally impaired; NK cells (CL:0000623) show impaired degranulation during HLH.

Subcellular level. The pathology is centered in the **cytoplasm (GO:0005737)**, where mutant CARD11 forms aggregates (mCADS) and assembles the CBM signalosome; signaling culminates in the **nucleus (GO:0005634)** via NF-κB translocation.

Localization / lateralization. Systemic and **bilateral** (generalized lymphoproliferation, splenomegaly). SPTCL presents as subcutaneous skin lesions.

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8. Temporal Development

- **Onset. Congenital / early childhood** — lymphocytosis is frequently detectable from infancy; "congenital B-cell lymphocytosis" is a descriptive term for the disease. Onset pattern is **chronic/insidious**.
- **Progression. Variable**: often stable and indolent over years; a subset progresses rapidly in infancy to HLH/fatal infection. Not relapsing-remitting in the classic sense, though EBV and infections cause episodic exacerbations.
- **Duration. Chronic, lifelong** (persistent lymphocytosis).
- **Remission.** No spontaneous resolution of the genetic lesion; symptoms can improve with immunomodulation (treatment-induced improvement).
- **Critical period.** The **first ~3 years of life** are the window of highest mortality — all six deaths in the 34-case review occurred before age 3 (PMID: 39998705), making early diagnosis and vigilance against HLH/infection the key intervention window.

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9. Inheritance and Population

Epidemiology. Ultra-rare: fewer than ~40 patients reported worldwide as of 2025 (34 cases in Nishinosono review PMID: 39998705; 29 in the Baldini-type review PMID: 40157432). **No formal prevalence or incidence estimate exists.**

Inheritance. Autosomal dominant, with both **de novo** and **inherited (mother→child)** cases documented. **One biallelic (homozygous R331P)** case is reported. Stated plainly (PMID: 40157432):

> *"Gain-of-function (GOF) mutations in CARD11 cause BENTA disease (B cell Expansion with NF-κB and T cell Anergy), an autosomal dominant disorder."*

- **Penetrance:** High but with **variable expressivity** — B-lymphocytosis is frequent but "not universally present."
- **Expressivity:** Highly variable (mild C49Y/K215del vs severe G123D/M365K/R331P).
- **Anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism / founder effects / consanguinity:** No founder effect established; biallelic R331P implies possible consanguinity in that pedigree. No documented germline mosaicism.
- **Carrier frequency:** Vanishingly low (GOF variants near-absent in gnomAD).

Population demographics. Reported cases span **multiple ethnicities — USA, Europe, India, China, Japan** — with no geographic clustering. **Both sexes affected**; no strong sex bias documented. Age distribution skews to congenital/pediatric diagnosis.

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10. Diagnostics

Laboratory tests. CBC with differential showing **persistent polyclonal B lymphocytosis** (20,000–90,000/mm³ in severe cases); flow cytometry demonstrating **polyclonal, naïve-predominant (CD19⁺) B-cell expansion** — distinguishing BENTA from monoclonal leukemias (e.g., CLL). Immunoglobulin levels variable; vaccine responses often preserved in mild cases.

Biomarkers. No validated circulating biomarker beyond the lymphocyte phenotype; HLH work-up (ferritin, soluble IL-2R, triglycerides, fibrinogen, NK-cell degranulation) when hyperinflammation is suspected.

Functional / immunologic assays. In-vitro NF-κB reporter assays and lymphocyte proliferation studies (reduced proliferation to TCR/mitogen stimulation = T-cell anergy); **mCADS aggregation assays** in research settings.

Genetic testing — definitive. Diagnosis is confirmed by **identification of a germline GOF CARD11 variant**. Recommended approach: **single-gene testing, targeted primary-immunodeficiency/inborn-errors-of-immunity gene panels, or WES/WGS**. Functional confirmation (NF-κB activation assay) supports VUS reclassification. Chromosomal microarray, karyotyping, FISH, mtDNA, and repeat-expansion testing are **not applicable**.

Clinical criteria / differential diagnosis. No formal consensus criteria. Differential includes: **monoclonal B-cell lymphocytosis / CLL** (BENTA is polyclonal, congenital), **other congenital lymphocytoses**, **ALPS**, **primary HLH**, and reactive EBV lymphocytosis. BENTA-HLH overlap is an important diagnostic pitfall (PMID: 37401643).

Screening. No newborn or population screening exists. **Cascade genetic testing** of relatives is appropriate once a familial variant is identified.

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11. Outcome / Prognosis

Prognosis is bimodal. Many patients are indolent; a minority die young. In the 34-case systematic review (PMID: 39998705):

> *"Among these cases, 15 were either asymptomatic or showed improvement without requiring any specific treatment. However, all six reported deaths were diagnosed before the age of 3 years, with two attributed to refractory hemophagocytic syndrome and four caused by opportunistic infections."*

Outcome metric	Finding
Asymptomatic / improved without treatment	15 / 34 (~44%)
Deaths	6 / 34 (~18%), **all before age 3**
Cause of death	HLH (2), opportunistic infection (4)
HLH complication rate	~18.8%
Malignancy	Elevated lifetime risk (B-cell lymphoma; SPTCL in 2)

Morbidity. Recurrent infections, splenomegaly-related issues, lymphoid hyperplasia (airway obstruction), HLH, and malignancy surveillance burden.

Prognostic factors. Severity correlates with the **specific allele** (mild C49Y/K215del vs severe G123D/M365K/R331P), **EBV exposure**, and **early HLH**. No validated prognostic biomarker panel exists. No formal 5-/10-year survival statistics are published given rarity.

12. Treatment

No curative or disease-specific approved therapy exists. Management is **supportive and immunomodulatory**, guided by severity.

Intervention	Role	NCIT suggestion	Evidence
Corticosteroids (prednisolone, dexamethasone)	First-line for symptomatic lymphoproliferation / HLH	NCIT:C312 (Prednisolone)	39998705, 36129242
Sirolimus (rapamycin, mTOR inhibitor)	Successful control of lymphoproliferation	NCIT:C1212 (Sirolimus)	39998705
Tacrolimus (calcineurin inhibitor)	Used with prednisolone in familial G123D case	NCIT:C1133 (Tacrolimus)	36129242
Etoposide (+ dexamethasone, HLH protocol)	For HLH episodes	NCIT:C491 (Etoposide)	37401643
Antibiotics / anti-infectives	Control of infections precipitating HLH	—	37401643
Adenoidectomy / tonsillectomy	Relief of airway obstruction from lymphoid hyperplasia	NCIT:C157808	reviews
HSCT	Considered for severe/ refractory cases (not established)	NCIT:C15431	reviews
Splenectomy — AVOID	**Worsens lymphocytosis**	—	25352053

Sirolimus success is explicitly documented (PMID: 39998705): *"Her symptoms improved with prednisolone and sirolimus."*

Mechanism-based / investigational. Because BENTA is driven by constitutive CBM→NF-κB signaling, **downstream-targeted agents are rational**: - **MALT1 paracaspase inhibitors** — developed for ABC-DLBCL, where *"B cell receptor-addicted lymphomas rely strongly on this cancer cell-intrinsic MALT1 protease function"* (PMID: 37126937; see also PMID: 34783281, PMID: 19897720). - **BTK inhibitors** (upstream of CBM in BCR signaling). - **SMAC mimetics** targeting cIAP1/2 in the CBM complex (PMID: 27070702).

None are yet standard for BENTA. No specific NCT-registered BENTA trials are established.

Pharmacogenomics. Not specifically characterized for BENTA.

13. Prevention

- **Primary prevention:** Not possible (germline genetic disease). **Genetic counseling** for affected families (autosomal dominant; 50% transmission risk from an affected heterozygous parent) is the main preventive tool, with optional **prenatal / preimplantation genetic testing** once a familial variant is known.
- **Secondary prevention:** Early molecular diagnosis and **cascade testing** of relatives; close surveillance in infancy for HLH and infection (the critical <3-year window).
- **Tertiary prevention:** Aggressive infection control, **avoidance of splenectomy**, EBV monitoring, and malignancy surveillance to prevent complications.
- **Immunization / prophylaxis:** Standard vaccination (many patients respond to conjugate vaccines); infection prophylaxis as clinically indicated. No disease-specific vaccine.
- **Public health / environmental:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy / ortholog:** Human *CARD11* (NCBI Gene 84433); mouse *Card11* ortholog (**NCBI Gene 108723**). Highly conserved across mammals.
- **Natural disease in other species:** No naturally occurring BENTA-equivalent disease is documented in companion animals or wildlife (no OMIA entry for a CARD11-GOF animal disease). The disease is known only from humans and engineered models.
- **Comparative biology:** The CBM→NF-κB axis is evolutionarily conserved; mouse *Card11* alleles recapitulate both GOF (activation) and LOF (immunodeficiency) ends of the human allelic series, confirming conserved mechanism.
- **Zoonotic / transmission:** Not applicable (non-infectious genetic disease).

15. Model Organisms

Mouse models (primary):

Model	Type	Phenotype	PMID
Card11 M365K knock-in	Germline GOF knock-in	Increased B- and T-lymphocyte activation and proliferation after antigen-receptor stimulation; models BENTA/DLBCL GOF spectrum	36960072
Card11 R30W	Point mutant (allele study)	Demonstrates CARD11 role in NF-κB, JNK, mTOR activation	34341167
Card11 knockout / "unmodulated" (LOF)	Null / LOF	Combined immunodeficiency; impaired B/T antigen-receptor-induced NF-κB — the **opposite** pole of the allelic series	—

In-vitro / cellular models (heavily used): - **HEK293T transfection** NF-κB reporter assays to quantify constitutive activation. - **Patient-derived primary lymphocytes** and lymphocyte cell lines (e.g., **Jurkat**) for signaling and mCADS aggregation studies (PMID: 32473470, PMID: 23129749, PMID: 40157432). - ABC-DLBCL cell lines carrying analogous somatic CARD11 mutations as a disease-adjacent model.

Phenotype recapitulation. The M365K knock-in reproduces the **lymphocyte hyperactivation/proliferation** core of BENTA. **Limitations:** mouse models do not fully capture human T-cell anergy, the HLH/EBV-driven mortality, or the human malignancy spectrum (EBV does not infect mice); the biallelic R331P human phenotype has no mouse equivalent.

Resources: MGI (mouse *Card11*), IMPC/IMSR for knockout lines; Cellosaurus/ATCC for cell lines.

Key Findings (Consolidated)

Finding 1 — BENTA is caused by germline gain-of-function *CARD11* mutations (autosomal dominant)

Snow et al. (2012) identified germline heterozygous **E127G** (familial) and **G116S** (unrelated) in the CARD11 coiled-coil domain, both driving constitutive NF-κB activation (PMID: 23129749). An expanding allelic series now includes G123D, C49Y, M365K, D357E, K215del, G123S, and the homozygous R331P. Inheritance is autosomal dominant (de novo and inherited), with one biallelic case (PMID: 40157432).

Finding 2 — Mechanism: GOF CARD11 self-aggregates to nucleate a constitutive CBM signalosome

Mutant CARD11 forms cytosolic aggregates (mCADS) colocalizing with MALT1 and phospho-IKK, driving antigen-independent NF-κB activity (PMID: 32473470). The CBM complex is the essential link between antigen receptors and NF-κB (PMID: 25087226).

Finding 3 — Clinical phenotype: early-onset polyclonal B-cell lymphocytosis with HLH and lymphoma risk

Core presentation: polyclonal B lymphocytosis, splenomegaly, lymphadenopathy, recurrent infections (PMID: 40157432). HLH in ~18.8%; lymphocytosis not universal. Severe cases reach 20,000–90,000/mm^3, exacerbated by EBV and splenectomy (PMID: 25352053).

Finding 4 — Treatment/prognosis: often indolent but early-childhood fatal complications

15/34 asymptomatic or improved without treatment; all 6 deaths before age 3 (2 HLH, 4 opportunistic infection) (PMID: 39998705). Corticosteroids, sirolimus, tacrolimus used; MALT1/BTK inhibition investigational.

Finding 5 — Model organisms: Card11 M365K knock-in mouse recapitulates GOF; Card11-null is immunodeficient

The M365K knock-in increases lymphocyte activation/proliferation (PMID: 36960072); Card11 KO causes combined immunodeficiency (opposite pole). In-vitro HEK293T and patient-cell assays are mainstays.

Finding 6 — CARD11 allelic series, epidemiology, identifiers

Three diseases from one gene (CID 615206 LOF; CADINS 617638 DN-LOF; BENTA 616452 GOF). Ultra-rare (<~40 patients). GOF variants near-absent in gnomAD.

Finding 7 — B-cell immunophenotype: polyclonal, naïve-surface, aberrant transcriptional program

Biallelic R331P patient showed massive naïve B-cell expansion with aberrant transcription (PMID: 41456721); B cells show upregulated cell-cycle genes and enhanced proliferation/survival (PMID: 25352053). T cells retain Vβ diversity but are hyporesponsive (PMID: 37401643).

Finding 8 — Predisposition to lymphoid malignancy (B-cell and T-cell)

The BENTA lesion class occurs somatically in ABC-DLBCL (PMID: 32473470); activating CARD11 mutations predispose to B-cell malignancy (PMID: 23129749); two unrelated patients developed SPTCL T-cell lymphoma (PMID: 37562721).

Mechanistic Model / Interpretation

BENTA is best understood as the **germline, whole-organism manifestation of the same oncogenic lesion that, acquired somatically, produces ABC-DLBCL**. A single structural change in CARD11 relieves its autoinhibition, allowing it to aggregate and self-assemble the CBM signalosome in the absence of antigen. The downstream NF-κB output is cell-type-divergent: B cells interpret constitutive NF-κB as a proliferation/survival signal and expand polyclonally, whereas T cells become anergic — a paradox likely reflecting chronic-signaling-induced desensitization and lineage-specific wiring. This divergence is the disease's defining signature ("B-cell Expansion" + "T-cell Anergy").

The **bimodal prognosis** maps onto allele strength and environmental triggers. Strong alleles plus EBV plus the vulnerable infant immune system can tip the system from indolent lymphoproliferation into HLH or fatal opportunistic infection; weak alleles produce lifelong but benign lymphocytosis. This model explains why **splenectomy worsens disease** (removing a filter redistributes/boosts the expansion), why **sirolimus/mTOR and calcineurin inhibition**

help (dampening the proliferative arm), and why **downstream MALT1/BTK inhibitors are the most rational targeted strategy** — they attack the single constitutively active node common to all BENTA alleles.

Evidence Base

PMID	Title (abbrev.)	Supports
23129749	*Congenital B-cell lymphocytosis explained by novel germline CARD11 mutations*	Causal gene, E127G/G116S, B-malignancy predisposition
32473470	*GOF mutations in CARD11 promote enhanced aggregation…*	mCADS aggregation mechanism, DLBCL link
25087226	*The CBM signalosome complex…*	CBM as antigen-receptor→NF-κB link
40157432	*BENTA disease or CARD11 GOF?*	AD inheritance, core phenotype, HLH 18.8%
25352053	*Germline CARD11 mutation… severe congenital B-cell lymphocytosis*	Severity, EBV/splenectomy, B-cell program
39998705	*Successful treatment with sirolimus…*	Outcomes (6 deaths <3y), sirolimus/prednisolone
41456721	*B-cell lymphocytosis… biallelic CARD11*	Naïve B-cell expansion, transcriptional reprogramming, biallelic R331P
37401643	*BENTA overlapping with HLH*	T-cell hyporesponsiveness, HLH overlap
37562721	*SPTCL in two BENTA individuals*	T-cell lymphoma predisposition
36129242	*Familial G123D…*	Inheritance, autoimmune features, tacrolimus
31897776	*K215del deletion…*	Mild allele, subtle NF-κB increase
25930198	*Mild lymphocytosis with C49Y*	Mild allele, variable expressivity
37126937 / 34783281 / 19897720 / 27070702	MALT1/CBM targeting in DLBCL	Investigational therapeutic rationale
34341167	Mouse Card11 R30W allele	CARD11 → NF-κB/JNK/mTOR; allelic series

All eight findings are **human-clinical** at their core, reinforced by **in-vitro functional** assays (NF-κB reporters, mCADS) and **mouse-model** corroboration (M365K GOF, Card11-null LOF).

Limitations and Knowledge Gaps

1. **Rarity limits epidemiology.** With <40 patients, there is no prevalence/incidence, no formal survival statistics, and selection/publication bias toward severe cases.
2. **Mechanism of T-cell anergy is incompletely resolved** — the paradox of an activating lesion causing T-cell hyporesponsiveness is inferred, not fully demonstrated.
3. **No genotype–phenotype rule is validated.** Allele-severity correlations are observational; the same variant can give variable expressivity.
4. **No disease-specific clinical trials or approved targeted therapy.** MALT1/BTK inhibition is extrapolated from DLBCL, not tested in BENTA.
5. **Model limitations.** Mice don't capture EBV-driven exacerbation, HLH mortality, or human malignancy spectrum; no model of biallelic R331P.
6. **No epigenetic or multi-omic disease-wide datasets**, and no QoL instruments applied.

Proposed Follow-up Experiments / Actions

1. **Establish an international BENTA registry** to capture natural history, genotype–phenotype correlations, and outcomes prospectively.
2. **Allele-resolved functional screen** of all reported CARD11 variants in a uniform NF-κB/mCADS assay to build a quantitative severity ladder.
3. **Preclinical test of MALT1 paracaspase inhibitors and BTK inhibitors** in the Card11 M365K knock-in mouse and patient-derived B cells, measuring lymphocytosis, NF-κB output, and HLH susceptibility.
4. **Single-cell multi-omics (scRNA-seq + CITE-seq) of paired B and T compartments** across mild vs severe patients to dissect the proliferation-vs-anergy divergence.
5. **Define EBV interaction mechanistically** — controlled infection of patient B cells / humanized models to clarify the EBV × CARD11-GOF HLH/lymphoma axis.

6. **Develop consensus diagnostic and surveillance guidelines**, emphasizing the <3-year critical window, splenectomy avoidance, and malignancy monitoring.
7. **Pilot sirolimus/mTOR vs calcineurin-inhibitor comparison** in symptomatic patients within a registry-embedded study.

Report compiled from 21 primary papers across a 5-iteration autonomous investigation. Evidence types are distinguished as human-clinical, in-vitro functional, and model-organism throughout. All quoted abstract excerpts are verbatim from the cited PMIDs.
